KEYNOTE-867: A Phase III, Randomized, Double-Blind Study of Stereotactic Body Radiotherapy With Pembrolizumab for Unresected Stage I or II Non–Small Cell Lung Cancer

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ABSTRACT

PURPOSE Stereotactic body radiotherapy (SBRT) is standard of care for inoperable stage I or II non–small cell lung cancer (NSCLC). KEYNOTE-867 (ClinicalTrials.gov identifier: NCT03924869) evaluated the addition of pembrolizumab to SBRT in participants diagnosed with stage I or II NSCLC that was inoperable or in participants who refused surgery.

METHODS Adults with untreated, stage I or II (T1 to limited T3, N0, M0) NSCLC (AJCC 8th edition) that was inoperable, or adults who refused surgery, were randomly assigned 1:1 to thoracic SBRT to primary tumors for ≤2 weeks concomitantly with pembrolizumab 200 mg or placebo once every 3 weeks for ≤17 cycles. The primary end point was event-free survival (EFS) by blinded independent central review.

RESULTS A total of 448 participants were randomly assigned to SBRT plus pembrolizumab (n = 227) or SBRT plus placebo (n = 221). The median time from random assignment to the data cutoff date (June 11, 2024) for the prespecified first interim analysis was 20.6 (range, 1.3-56.8) months. The median EFS (95% CI) was 31.2 (22.5 to 39.1) months and 28.3 (19.8 to 33.1) months for the pembrolizumab and placebo arms, respectively (hazard ratio, 0.92 [95% CI, 0.69 to 1.24]; $P = .2932$). Among participants who received ≥1 dose of study treatment, grade 3-5 treatment-related adverse events (AEs) occurred in 46/226 (20.4%) participants in the pembrolizumab arm and 8/218 (3.7%) participants in the placebo arm. There were five treatment-related deaths all of which occurred in the pembrolizumab arm (2.2%).

CONCLUSION The addition of pembrolizumab to SBRT did not improve EFS and was associated with a higher incidence of treatment-related AEs compared with SBRT alone in participants with inoperable stage I or II NSCLC. Based on the benefit-risk profile, the KEYNOTE-867 trial was prematurely stopped, and all study treatment was discontinued.

ACCOMPANYING CONTENT

- Graphical Abstract
- Appendix
- Data Sharing Statement
- Protocol

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INTRODUCTION

Standard treatment for patients with stage I or II non–small cell lung cancer (NSCLC) is surgical resection, but for patients who are medically inoperable or who refuse surgery, stereotactic body radiotherapy (SBRT) is the treatment of choice.[1] Studies have shown SBRT to be highly effective for primary tumor control in participants with early-stage NSCLC who do not undergo surgery, although rates of recurrence after SBRT, including distant recurrences, are considerable.[2-6] Thus, systemic therapy in addition to SBRT is needed to improve outcomes for patients who do not undergo surgery for early-stage NSCLC.

Preclinical evidence shows that radiotherapy can boost antitumor immune responses by increasing antigen production and presentation and by stimulating dendritic cells to engulf tumor cells and to prime T cells.[7-11] Previous clinical studies of immune checkpoint inhibitors used after radiotherapy or concurrent with chemoradiotherapy have

CONTEXT

Key Objective

How do the efficacy and safety of stereotactic body radiotherapy (SBRT) with pembrolizumab compare with those of SBRT with placebo for the treatment of participants with unresected stage I or II non–small cell lung cancer (NSCLC)?

Knowledge Generated

The addition of pembrolizumab to SBRT did not improve event-free survival compared with SBRT plus placebo. The incidence of treatment-related adverse events, including those grades 3-5, was higher in the pembrolizumab arm than the placebo arm.

Relevance *(T.E. Stinchcombe)*

Patients with stage I and II NSCLC who undergo SBRT should not receive immunotherapy.*

*Relevance section written by *JCO* Associate Editor Thomas E. Stinchcombe, MD, FASCO.

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reported favorable outcomes in participants with locally advanced or metastatic NSCLC.[12-15] In a secondary analysis of the phase I KEYNOTE-001 study, both progression-free survival (PFS) and overall survival (OS) with pembrolizumab were longer for participants with advanced NSCLC who had received prior radiotherapy than for those who had not.[12] Furthermore, among participants with newly diagnosed early-stage NSCLC (N = 141 received treatment), a phase II study of SBRT with or without concomitant immunotherapy found that the combination improved 4-year event-free survival (EFS) with manageable toxicity,[16] suggesting a possible role for immunotherapy to be added to SBRT in the treatment of early-stage NSCLC.

Here, we report the findings of KEYNOTE-867, a phase III, randomized, controlled, multisite study of SBRT with concomitant pembrolizumab or placebo in participants with unresected stage I or II NSCLC.

METHODS

Study Participants

Participants were adults (≥18 years) with untreated, unresected, cytologically or histologically confirmed stage I or II (T1 to limited T3, N0, M0) NSCLC (per the American Joint Committee on Cancer staging system, 8th edition[17]) that was inoperable (as determined by the site's multidisciplinary tumor board or site investigator) or who had a documented unwillingness to undergo surgical resection for stage I or II NSCLC. Other key eligibility criteria included an ability to receive SBRT, an Eastern Cooperative Oncology Group (ECOG) performance status score of 0, 1, or 2, and adequate organ function. Presence of an ultra-centrally located tumor abutting or invading the trachea, esophagus, or bronchial tree was an exclusion criterion. Investigators obtained informed consent from each participant or each participant's guardian. Full eligibility criteria are included in the study protocol (Protocol, online only).

Study Design and Treatments

KEYNOTE-867 (ClinicalTrials.gov identifier: NCT03924869) enrolled participants at 193 study sites worldwide (Appendix Table A1, online only). Participants were randomly assigned 1:1 to SBRT plus pembrolizumab 200 mg or SBRT plus placebo, administered by intravenous infusion once every 3 weeks for up to 17 cycles. SBRT regimens were administered over a period of no more than 2 weeks, beginning on day 1 of cycle 1 following the administration of pembrolizumab or placebo. Participants with peripheral tumors (defined as tumors located within the lung parenchyma with the gross tumor volume outside a 2-cm zone of the proximal bronchial tree) received 45-60 Gy in three fractions (preferred), 48-50 Gy in four fractions, or 50-55 Gy in five fractions. Participants with tumors abutting the chest wall (defined as tumors with planning target volumes that result in 1 cm or more of chest wall contact) received 48-50 Gy in four fractions or 50-55 Gy in five fractions. Participants with central tumors (defined as tumors within 2 cm of the proximal bronchial tree and/or abutting the mediastinal pleura) received 50-55 Gy in five fractions or 60-70 Gy in eight fractions. Random assignment was stratified by disease stage (I *v* II), ECOG performance status (0 or 1 *v* 2), geographic region (east Asia *v* non-east Asia), and reason for not receiving surgery (medically inoperable *v* refused surgery). The study protocol and amendments were reviewed and approved by the appropriate institutional review board at each study site. The study was conducted in compliance with global standards, local and/or national regulations of each study site, and International Council for Harmonisation Good Clinical Practice guidelines, and in accordance with the ethical principles that originated with the Declaration of Helsinki.

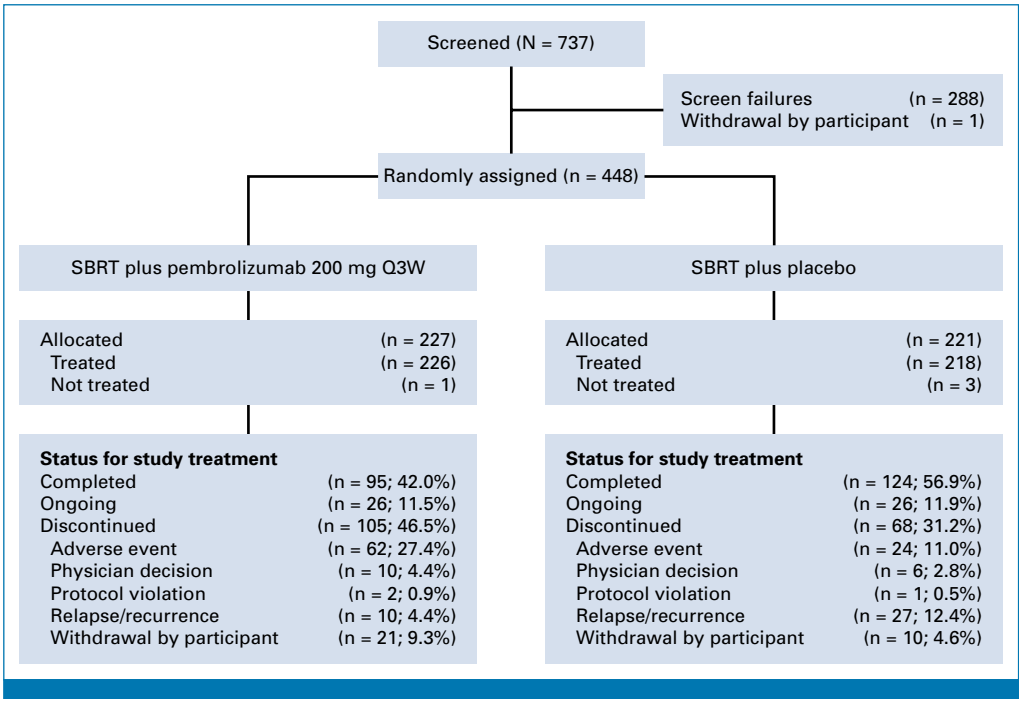


FIG 1. CONSORT diagram. SBRT, stereotactic body radiotherapy; Q3W,once every 3 weeks.

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End Points and Assessments

The primary end point was EFS (defined as the time from random assignment to local, regional, or distant recurrence of NSCLC or death from any cause). Secondary efficacy end points were OS (defined as the time from random assignment to death from any cause) and time from random assignment to distant metastases or death from any cause, whichever occurred first. A tertiary end point was disease-specific survival (DSS; defined as the time from random assignment to death due to NSCLC). A whole-body fluorodeoxyglucose-positron emission tomography scan was performed during screening. Computed tomography scans of the chest and abdomen were performed at 12 weeks post-random assignment (or sooner if clinically indicated), every 16 weeks until 156 weeks, and every 6 months thereafter. Response was assessed by blinded independent central review per RECIST v1.1.

For the secondary end point of safety and tolerability, all adverse events (AEs) from random assignment through 30 days and serious AEs through 90 days after the last administration of study treatment were collected. AEs were assessed by the National Cancer Institute Common Terminology Criteria for Adverse Events (CTCAE) v4.03.

Another secondary end point was the change from baseline scores in the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ-C30 items 1-5 (physical functioning) and items 29-30 (global health status/quality of life), and the EORTC QLQ-LC13 item 1 (cough), item 8 (dyspnea), and item 10 (chest pain). Questionnaires were administered in the local language and were completed electronically by participants before dosing at cycles 1, 5, 9, 13, and 17 during treatment, at the treatment discontinuation visit, and at the 30-day safety follow-up visit.

Statistical Analyses

Efficacy analyses were conducted in the intention-to-treat population of all randomly assigned participants. The nonparametric Kaplan-Meier method was used to estimate EFS and OS in each treatment group, and the treatment differences were assessed by the stratified log-rank test. The hazard ratios (HRs) and 95% CIs were determined using a stratified Cox proportional hazard model with a single treatment covariate. The stratification factors used for random assignment were applied to both the stratified log-rank test and the stratified Cox model. Multiplicity was strongly controlled at a one-sided alpha of .025 across the EFS and OS hypotheses and among the interim and final analyses using the graphical method of Maurer and Bretz.[18] All alpha was allocated to testing the superiority of SBRT plus pembrolizumab versus SBRT plus placebo for EFS. If the EFS null hypothesis was rejected, the full alpha would be allocated to test the OS hypothesis. Three interim analyses and a final analysis were planned. The first interim analysis was to be performed after approximately 161 EFS events had accrued and approximately 4 months had elapsed after the last participant was randomly assigned. After analysis by the independent Data Monitoring Committee and subsequent review by the sponsor, the study was stopped for futility, and first interim analysis became the final analysis (data cutoff date June 11, 2024). Efficacy analyses were reported at the

TABLE 1. Baseline Participant Demographics and Clinical Characteristics

Characteristic	SBRT Plus Pembrolizumab (n = 227)	SBRT Plus Placebo (n = 221)
Age, years, median (range)	73.0 (50-93)	73.0 (49-87)
<65 years, No. (%)	33 (14.5)	33 (14.9)
≥65 years, No. (%)	194 (85.5)	188 (85.1)
Sex, No. (%)		
Male	138 (60.8)	121 (54.8)
Female	89 (39.2)	100 (45.2)
Race, No. (%)		
American Indian or Alaska Native	0 (0)	1 (0.5)
Asian	31 (13.7)	26 (11.8)
Black or African American	4 (1.8)	6 (2.7)
Multiple	0 (0)	2 (0.9)
White	183 (80.6)	177 (80.1)
Missing	9 (4.0)	9 (4.1)
Region, No. (%)		
Europe	97 (42.7)	106 (48.0)
North America	45 (19.8)	41 (18.6)
East Asia	29 (12.8)	26 (11.8)
Latin America	22 (9.7)	18 (8.1)
Rest of the world	34 (15.0)	30 (13.6)
ECOG performance status, No. (%)		
0	86 (37.9)	84 (38.0)
1	123 (54.2)	122 (55.2)
2	18 (7.9)	15 (6.8)
Histology, No. (%)		
Squamous	83 (36.6)	81 (36.7)
Nonsquamous	144 (63.4)	140 (63.3)
Stage of disease, No. (%)		
IA1	19 (8.4)	13 (5.9)
IA2	67 (29.5)	77 (34.8)
IA3	72 (31.7)	58 (26.2)
IB	46 (20.3)	49 (22.2)
IIA	19 (8.4)	19 (8.6)
IIB	4 (1.8)	5 (2.3)
Smoking status, No. (%)		
Never	21 (9.3)	18 (8.1)
Ever	206 (90.7)	203 (91.9)
Tumor location, No. (%)		
Abutting the chest wall	105 (46.3)	105 (47.5)
Peripheral	88 (38.8)	87 (39.4)
Central	33 (14.5)	25 (11.3)
Missing	1 (0.4)	4 (1.8)
Reason for not receiving surgery, No. (%)		
Refused surgery	43 (18.9)	36 (16.3)
Medically inoperable	184 (81.1)	185 (83.7)

Abbreviations: ECOG, Eastern Cooperative Oncology Group; SBRT, stereotactic body radiotherapy.

time of the interim/final analysis, except for the DSS analysis, which was based on the end-of-study analysis (data cutoff date January 20, 2025).

Safety analyses were conducted in all randomly assigned participants who received at least one dose of study treatment. Safety findings were reported at the time of the interim/final analysis (data cutoff date June 11, 2024).

Participant-reported outcomes (PRO) analyses were conducted in the population of participants who received at least one dose of study treatment and had at least one PRO assessment available. A constrained longitudinal data analysis model was applied for each end point, with the PRO score as the response variable, and treatment, time, treatment by time interaction, and clinical study stratification factors as covariates. Least-squares mean change from baseline were summarized for each outcome. The difference in the least-squares mean change from baseline was reported at week 24. PRO findings were reported at the time of the interim/final analysis (data cutoff date June 11, 2024).

RESULTS

Participants and Treatments

Of 737 participants screened, 448 were randomly assigned from September 2, 2019, to January 3, 2024, to receive SBRT plus pembrolizumab (N = 227) or SBRT plus placebo (N = 221; Fig 1). Of the 289 participants not randomly assigned, the most common reasons were: they did not meet diagnostic criteria for inclusion (n = 121), did not provide written informed consent (n = 48), and were unable to receive SBRT (n = 41). The accrual goal of approximately 436 participants was met. The median time from random assignment to data cutoff on June 11, 2024, was 20.6 months (range, 1.3-56.8 months). Baseline participant demographics and clinical characteristics were generally balanced between treatment arms (Table 1). The median age was 73.0 years in the study population, with 14.7% of participants <65 years of age. Participants with tumors ≤2 cm in size accounted for 39.3% of the study population (disease stages IA1 and IA2). Most participants (82.4%) did not receive surgery because they were deemed medically inoperable.

All randomly assigned participants received study treatment, except for one participant in the pembrolizumab arm and three participants in the placebo arm. The median duration of pembrolizumab or placebo treatment was 9.7 months (range, 1.0 day-16.0 months) in the pembrolizumab arm and 11.1 months (range, 1.0 day-15.5 months) in the placebo arm, and the median number of cycles of pembrolizumab and placebo received was 13.5 (range, 1-17) and 17.0 (range, 1-17), respectively. In both the pembrolizumab and placebo arms, the most common SBRT regimen received was 50 Gy in five fractions, followed by 48 Gy in four fractions (Appendix

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Table A2). The median planning target volume was 35.0 cc (range, 6.0-298.0 cc) in the pembrolizumab arm and 32.0 cc (range, 3.0-233.0 cc) in the placebo arm.

Efficacy

With the occurrence of 182 (40.6%) events, the median EFS was 31.2 months (95% CI, 22.5 to 39.1) in the pembrolizumab arm and 28.3 months (95% CI, 19.8 to 33.1) in the placebo arm. EFS was not significantly improved in the pembrolizumab arm compared with the placebo arm (HR, 0.92 [95% CI, 0. 69 to 1.24]; P = .29; Fig 2A). In addition, OS was not improved in the pembrolizumab arm compared with the placebo arm (HR, 1.33 [95% CI, 0.93 to 1.90]; Fig 2B). The OS event rate was 30.4% in the pembrolizumab arm and 24.4% in the placebo arm. The median OS was 54.7 months (95% CI,

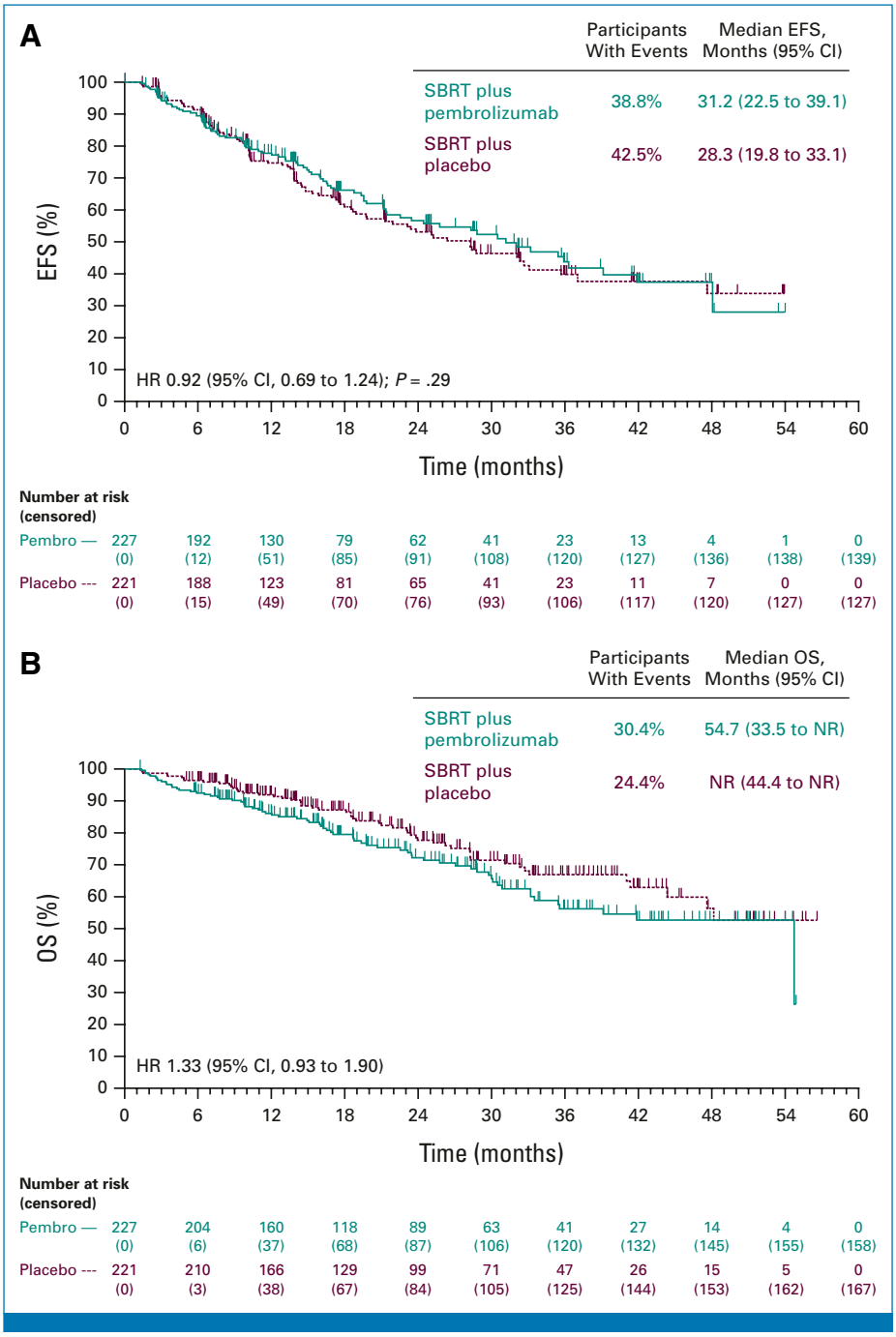


FIG 2. (A) Kaplan-Meier estimates of EFS based on blinded independent central review per RECIST v1.1 and (B) OS in the SBRT plus pembrolizumab arm (solid) and SBRT plus placebo arm (dashed). Data cutoff for these analyses was on June 11, 2024. EFS, event-free survival; HR, hazard ratio; OS, overall survival; SBRT, stereotactic body radiotherapy.

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33.5 months to not reached) in the pembrolizumab arm and not reached (95% CI, 44.4 months to not reached) in the placebo arm.

Subgroup analysis results for EFS and OS were generally consistent with the analysis results for EFS and OS in the overall study population (Fig 3). Results by age group suggested that adding pembrolizumab may have improved EFS and OS among participants younger than 65 years, but in both cases the 95% CIs were wide and overlapped with those of the overall analyses (Fig 3). Kaplan-Meier analyses of EFS and OS conducted post hoc also suggested differential outcomes by age group (Appendix Figs A1 and A2).

With 147 (32.8%) events, the median time to death or distant metastases was 39.1 months (95% CI, 30.5 months to not reached) in the pembrolizumab arm and 41.3 months (95% CI, 31.1 months to not reached) in the placebo arm. There was no difference in the risk of death or distant metastasis between treatment groups (HR, 1.02 [95% CI, 0.74 to 1.42]).

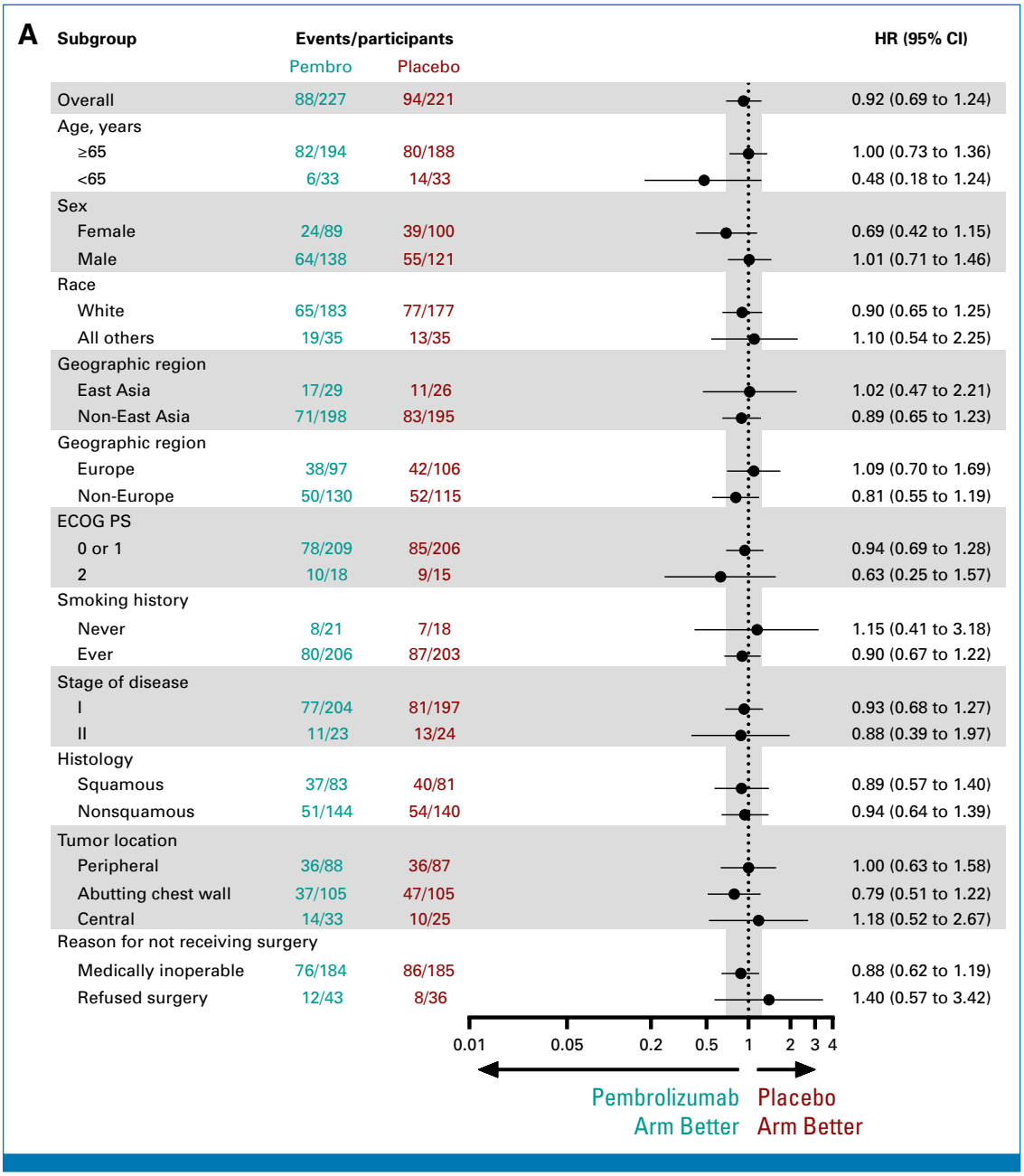


Fig 3. Forest plots of (A) EFS and (B) OS in subgroups. Data cutoff for these analyses was on June 11, 2024. ECOG PS, Eastern Cooperative Oncology Group performance status; EFS, event-free survival; HR, hazard ratio; NA, not applicable; OS, overall survival; pembro, SBRT plus pembrolizumab arm; placebo, SBRT plus placebo arm; SBRT, stereotactic body radiotherapy. (continued on following page)

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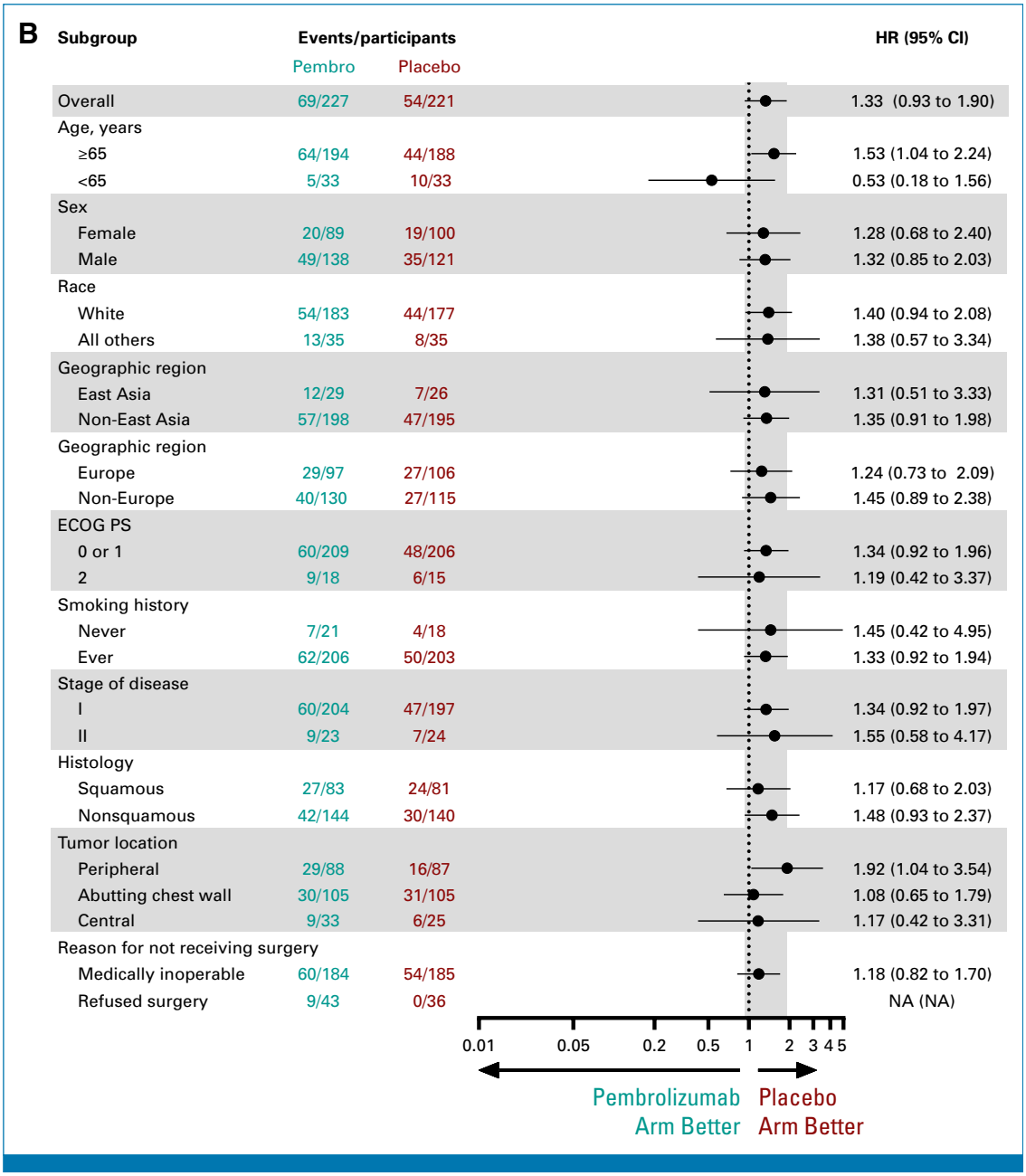


FIG 3. (Continued).

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At the data cutoff date of June 11, 2024, for the prespecified interim analysis, DSS data were not yet mature. Results of the DSS analysis performed using the data from the end-of-study analysis (data cutoff date of January 20, 2025) showed that DSS was not improved in the pembrolizumab arm compared with the placebo arm (HR, 1.44 [95% CI, 0.75 to 2.76]; Appendix Fig A3).

Safety

The incidence of treatment-related AEs was higher in the SBRT plus pembrolizumab arm compared with the SBRT plus placebo arm (73.5% *v* 50.0%; Table 2; Appendix Table A3). The treatment-related AEs occurring most frequently in the pembrolizumab versus placebo arms included hypothyroidism (13.7% *v* 2.8%), rash (12.4% *v* 5.0%), pruritus (11.9% *v* 5.5%), and fatigue (11.5% *v* 8.7%; Table 2). AEs leading to death occurred in 22 (9.7%) participants in the SBRT plus pembrolizumab arm, of which 5 (2.2%) events were considered related to treatment: pneumonia (n = 3), bronchopneumopathy (n = 1), and pneumonitis (n = 1; Table 2; Appendix Table A3). Fifteen (6.9%) participants in the SBRT plus placebo arm experienced an AE that led to death, but none of those events was considered treatment related (Appendix Table A3). The causes of death for all participants who died on study are listed in Appendix Table A4. The most

TABLE 2. Summary of Treatment-Related Adverse Events

Participants, No. (%)	SBRT Plus Pembrolizumab (n = 226)	SBRT Plus Placebo (n = 218)
With TRAEs[a]	166 (73.5)	109 (50.0)
Grade 3-5 TRAEs	46 (20.4)	8 (3.7)
Serious TRAEs	32 (14.2)	5 (2.3)
Died	5 (2.2)[b]	0 (0.0)
Discontinued treatment because of TRAE	36 (15.9)	5 (2.3)
Discontinued pembrolizumab or placebo	36 (15.9)	5 (2.3)
Discontinued radiation	0 (0.0)	0 (0.0)
Discontinued treatment because of serious TRAE	15 (6.6)	2 (0.9)
Discontinued pembrolizumab or placebo	15 (6.6)	2 (0.9)
Discontinued radiation	0 (0.0)	0 (0.0)

TRAEs With Incidence ≥5%	Grade 1 to 2, No. (%)	Grade 3 to 4, No. (%)	Grade 5, No. (%)	Grade 1 to 2, No. (%)	Grade 3 to 4, No. (%)	Grade 5, No. (%)
Hypothyroidism	31 (13.7)	0 (0)	0 (0)	6 (2.8)	0 (0)	0 (0)
Rash	28 (12.4)	0 (0)	0 (0)	11 (5.0)	0 (0)	0 (0)
Pruritus	26 (11.5)	1 (0.4)	0 (0)	12 (5.5)	0 (0)	0 (0)
Fatigue	26 (11.5)	0 (0)	0 (0)	19 (8.7)	0 (0)	0 (0)
Pneumonitis	17 (7.5)	4 (1.8)	1 (0.4)	4 (1.8)	0 (0)	0 (0)
Radiation pneumonitis	22 (9.7)	0 (0)	0 (0)	18 (8.3)	0 (0)	0 (0)
Diarrhea	17 (7.5)	1 (0.4)	0 (0)	16 (7.3)	0 (0)	0 (0)
Arthralgia	13 (5.8)	1 (0.4)	0 (0)	5 (2.3)	0 (0)	0 (0)
Hyperthyroidism	14 (6.2)	0 (0)	0 (0)	3 (1.4)	0 (0)	0 (0)

Abbreviations: SBRT, stereotactic body radiotherapy; TRAE, treatment-related adverse event.
[a]Determined by the investigator to be related to treatment.
[b]Includes three events of pneumonia, one event of bronchopneumopathy, and one event of pneumonitis.

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common cause of death in both treatment arms was malignant neoplasm progression.

The incidence of all-cause pneumonitis was higher in the SBRT plus pembrolizumab arm than the SBRT plus placebo arm (20.8% *v* 10.6%), but the incidence of radiation pneumonitis was similar between arms (9.7% *v* 9.2%; Appendix Table A5). Most events of pneumonitis and radiation pneumonitis were grade 1 or 2 in severity. Seven (3.1%) participants, all in the pembrolizumab arm, had grade 3-5 events of pneumonitis and radiation pneumonitis, including one participant who had grade 5 pneumonitis (Appendix Table A5). There were no unexpected immune-mediated AEs or infusion reactions reported (Appendix Table A6).

PRO Measures

Between baseline and week 24, global health status scores in both treatment arms decreased, with no differences between arms (Appendix Table A7). Physical and role functioning scores at week 24 also decreased from baseline in both treatment arms (Appendix Table A7). For all disease-related symptoms scales, the mean change from baseline in symptom scores remained stable at week 24 in both treatment arms with no clinically meaningful differences between the treatment arms (Appendix Table A7).

DISCUSSION

At the prespecified first interim analysis of the KEYNOTE-867 study, SBRT plus pembrolizumab did not significantly improve EFS compared with SBRT plus placebo. Because EFS was not statistically significant at the time of the analysis, the OS hypothesis was not formally tested. The point estimate of OS HR did not favor SBRT plus pembrolizumab over SBRT plus placebo. Although the AEs observed in the SBRT plus pembrolizumab arm were generally consistent with the established safety profiles of the individual treatments and the underlying disease, the addition of pembrolizumab to SBRT increased the overall frequency and severity of AEs. Treatment-related AEs, including those that were grade 3-5, serious, or resulted in discontinuation of study treatment, were more frequent in the pembrolizumab arm. Although the incidence of radiation pneumonitis was similar between treatment arms, all-cause pneumonitis occurred more frequently in the pembrolizumab arm. Based on the results of the interim analysis, the study was stopped, and all study treatments were discontinued.

Although there was no benefit for adding pembrolizumab to SBRT in the overall population, results of both subgroup and post hoc analyses suggested that SBRT plus pembrolizumab may improve both EFS and OS for participants younger than 65 years, although the number of participants <65 years was small (<15% of the study population). KEYNOTE-867 was not designed to assess the effect of study treatment on survival end points by age. Notably, a subgroup analysis in the study of SBRT plus nivolumab versus SBRT similarly showed improved EFS in the nivolumab arm among participants at or younger than the median age (≤72 years) but not among older participants.[16] Further investigations are needed to evaluate the potential utility of adding immunotherapy to SBRT in younger patients.

Despite the successes of (chemo)radiation therapy followed by immunotherapy in participants with locally advanced or metastatic NSCLC,[12-14] the benefit of concomitant (chemo) radiation therapy and immunotherapy for NSCLC has not been demonstrated. In stage III NSCLC, the phase III PACIFIC study of durvalumab or placebo after chemoradiation (N = 713) showed significantly improved PFS and OS with durvalumab consolidation therapy, but the phase III PACIFIC-2 study of durvalumab or placebo with concomitant chemoradiation (N = 328) found no differences in PFS and OS between treatment arms.[13,14,19] In early-stage NSCLC, the phase II study of SBRT with or without nivolumab reported significantly better 4-year EFS in the nivolumab arm,[16] whereas our phase III KEYNOTE-867 study found no significant improvement in EFS or OS with the addition of pembrolizumab to SBRT. Differences in study design between the SBRT plus nivolumab study and KEYNOTE-867 should be noted, including a shorter duration of immunotherapy treatment with nivolumab (4-7 cycles once every 4 weeks) and the inclusion of participants with recurrent lung cancer in the SBRT plus nivolumab study. Additionally, a higher proportion of participants in the SBRT plus nivolumab study had small-sized tumors (≤2 cm) at baseline (60.9% *v* 39.3% in KEYNOTE-867). It is also worth noting that a preliminary report of results from a phase III, randomized trial of SBRT plus atezolizumab versus SBRT in participants with high-risk early-stage NSCLC (N = 403) showed no differences in OS and PFS between the two treatment arms and a higher incidence of grade 3-5 AEs with atezolizumab.[20] Full results from this SBRT plus atezolizumab study, as well as the phase III PACIFIC-4 study of SBRT plus durvalumab or placebo,[21] should provide further insight into the overall clinical utility of treatment with immunotherapy added to SBRT for early-stage NSCLC.

In summary, interim analysis results of the KEYNOTE-867 study did not support the combination of SBRT plus pembrolizumab for the treatment of unresected stage I or II NSCLC. EFS and OS were not improved with the addition of pembrolizumab to SBRT, and the combination was associated with greater toxicity, compared with SBRT alone. Based on these results, the study was stopped early, and all study treatment was discontinued.

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AFFILIATIONS

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DATA SHARING STATEMENT

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REFERENCES

1. National Comprehensive Cancer Netwrok: NCCN Clinical Practice Guidelines in Oncology—Non-Small Cell Lung Cancer v7.2025, 2025. https://www.nccn.org
2. Baumann P, Nyman J, Hoyer M, et al: Outcome in a prospective phase II trial of medically inoperable stage I non-small-cell lung cancer patients treated with stereotactic body radiotherapy. J Clin Oncol 27:3290-3296, 2009
3. Crabtree TD, Denlinger CE, Meyers BF, et al: Stereotactic body radiation therapy versus surgical resection for stage I non-small cell lung cancer. J Thorac Cardiovasc Surg 140:377-386, 2010
4. Onishi H, Shirato H, Nagata Y, et al: Stereotactic body radiotherapy (SBRT) for operable stage I non-small-cell lung cancer: Can SBRT be comparable to surgery? Int J Radiat Oncol Biol Phys 81: 1352-1358, 2011
5. Timmerman RD, Paulus R, Pass HI, et al: Stereotactic body radiation therapy for operable early-stage lung cancer: Findings from the NRG Oncology RTOG 0618 trial. JAMA Oncol 4:1263-1266, 2018
6. Timmerman RD, Hu C, Michalski JM, et al: Long-term results of stereotactic body radiation therapy in medically inoperable stage I non-small cell lung cancer. JAMA Oncol 4:1287-1288, 2018
7. Apetoh L, Ghiringhelli F, Tesniere A, et al: Toll-like receptor 4-dependent contribution of the immune system to anticancer chemotherapy and radiotherapy. Nat Med 13:1050-1059, 2007
8. Aymeric L, Apetoh L, Ghiringhelli F, et al: Tumor cell death and ATP release prime dendritic cells and efficient anticancer immunity. Cancer Res 70:855-858, 2010
9. Lugade AA, Moran JP, Gerber SA, et al: Local radiation therapy of B16 melanoma tumors increases the generation of tumor antigen-specific effector cells that traffic to the tumor. J Immunol 174: 7516-7523, 2005
10. Gameiro SR, Jammeh ML, Wattenberg MM, et al: Radiation-induced immunogenic modulation of tumor enhances antigen processing and calreticulin exposure, resulting in enhanced T-cell killing. Oncotarget 5:403-416, 2014
11. Schaue D, Ratikan JA, Iwamoto KS, et al: Maximizing tumor immunity with fractionated radiation. Int J Radiat Oncol Biol Phys 83:1306-1310, 2012
12. Shaverdian N, Lisberg AE, Bornazyan K, et al: Previous radiotherapy and the clinical activity and toxicity of pembrolizumab in the treatment of non-small-cell lung cancer: A secondary analysis of the KEYNOTE-001 phase 1 trial. Lancet Oncol 18:895-903, 2017
13. Antonia SJ, Villegas A, Daniel D, et al: Durvalumab after chemoradiotherapy in stage III non-small-cell lung cancer. N Engl J Med 377:1919-1929, 2017
14. Antonia SJ, Villegas A, Daniel D, et al: Overall survival with durvalumab after chemoradiotherapy in stage III NSCLC. N Engl J Med 379:2342-2350, 2018
15. Jabbour SK, Lee KH, Frost N, et al: Pembrolizumab plus concurrent chemoradiation therapy in patients with unresectable, locally advanced, stage III non-small cell lung cancer: The phase 2 KEYNOTE-799 nonrandomized trial. JAMA Oncol 7:1-9, 2021
16. Chang JY, Lin SH, Dong W, et al: Stereotactic ablative radiotherapy with or without immunotherapy for early-stage or isolated lung parenchymal recurrent node-negative non-small-cell lung cancer: An open-label, randomised, phase 2 trial. Lancet 402:871-881, 2023
17. Detterbeck FC, Boffa DJ, Kim AW, et al: The eighth edition lung cancer stage classification. Chest 151:193-203, 2017
18. Maurer W, Glimm E, Bretz F: Multiple and repeated testing of primary, coprimary, and secondary hypotheses. Stat Biopharm Res 3:336-352, 2011
19. Bradley JD, Sugawara S, Lee KH, et al: Simultaneous durvalumab and platinum-based chemoradiotherapy in unresectable stage III non-small cell lung cancer: The phase III PACIFIC-2 study. J Clin Oncol 43:3610-3621, 2025
20. Simone CB II, Daly ME, Redman MW, et al: SWOG/NRG S1914: Randomized phase III trial of induction/consolidation atezolizumab + SBRT versus SBRT alone in high risk, early-stage NSCLC. J Clin Oncol 43, 2025 (suppl 16; abstr 8003)
21. Robinson CG, Xing L, Tanaka H, et al: Phase 3 study of durvalumab with SBRT for unresected stage I/II, lymph-node negative NSCLC (PACIFIC-4/RTOG3515). J Clin Oncol 41, 2023 (suppl 16; abstr TPS8607)

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

KEYNOTE-867: A Phase III, Randomized, Double-Blind Study of Stereotactic Body Radiotherapy With Pembrolizumab for Unresected Stage I or II Non–Small Cell Lung Cancer

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APPENDIX

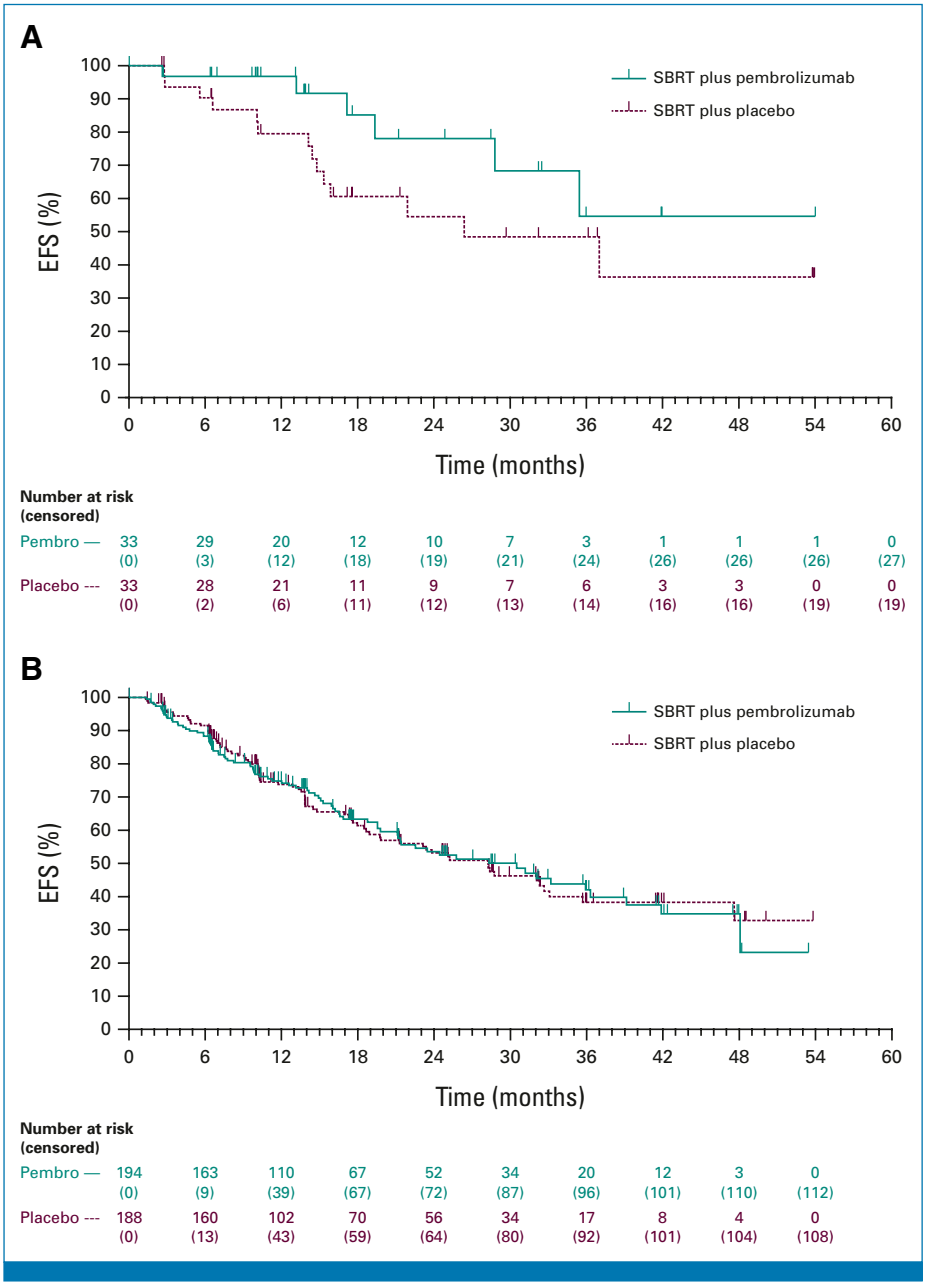


FIG A1. Kaplan-Meier estimates of EFS based on blinded independent central review per RECIST v1.1 for participants in the SBRT plus pembrolizumab arm (solid) and SBRT plus placebo arm (dashed) who were (A) <65 years and (B) ≥65 years. Data cutoff for these analyses was on June 11, 2024. EFS, event-free survival; SBRT, stereotactic body radiotherapy.

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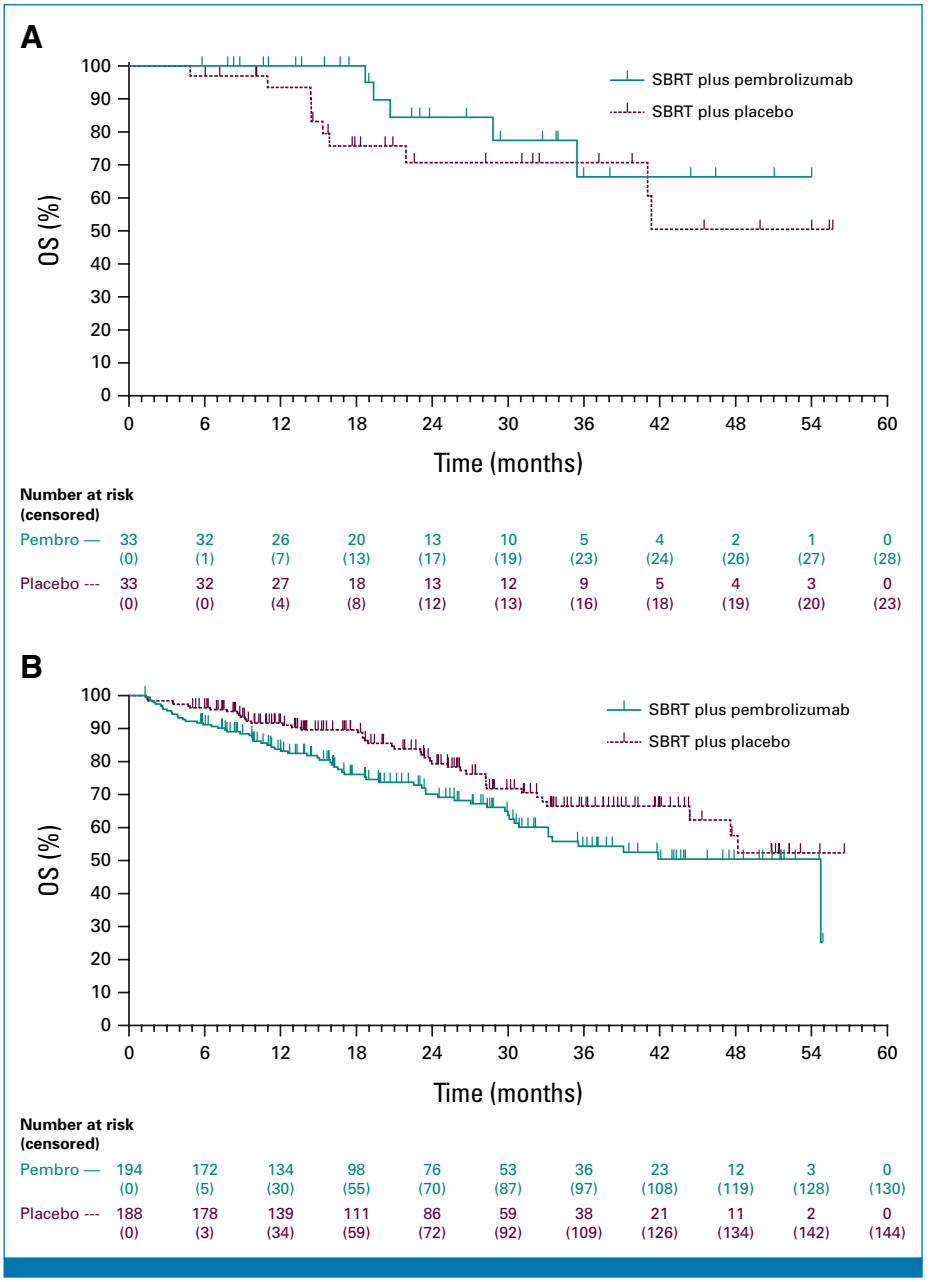


FIG A2. Kaplan-Meier estimates of OS for participants in the SBRT plus pembrolizumab arm (solid) and SBRT plus placebo arm (dashed) who were (A) <65 years and (B) ≥65 years. Data cutoff for these analyses was on June 11, 2024. OS, overall survival; SBRT, stereotactic body radiotherapy.

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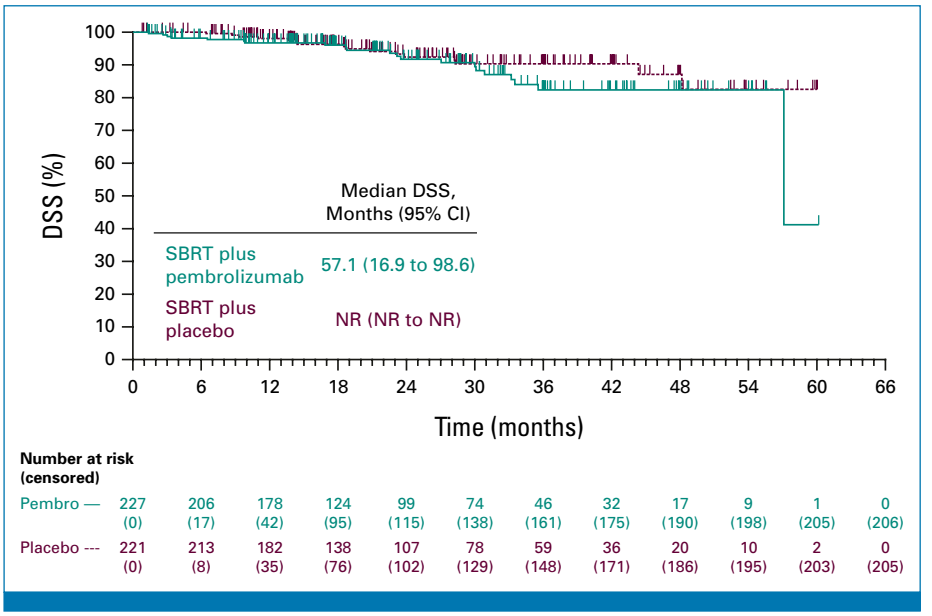


FIG A3. Kaplan-Meier estimates of DSS for participants in the SBRT plus pembrolizumab arm (solid) and SBRT plus placebo arm (dashed). Data cutoff for this analysis was on January 20, 2025. DSS, disease-specific survival; SBRT, stereotactic body radiotherapy.

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TABLE A1. List of Current and Former Principal Study Investigators and Sites That Randomized One or More Participants

Location	Current and Former Principal Investigator(s)	Site
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	Karina Vera	Hospital Britanico de Buenos Aires, Buenos Aires, Argentina
	Lorena Lupinacci	Hospital Italiano de Buenos Aires-Clinical Oncology, Buenos Aires, Argentina
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	Juan Cundom	Instituto de Diagnostico e Investigaciones Metabolicas, Buenos Aires, Argentina
Australia	Brett Gordon Maxwell Hughes	Royal Brisbane and Women's Hospital, Queensland, Australia
	Louise Nott	Icon Cancer Center Hobart, Tasmania, Australia
Austria	Sylvia Hartl; Marie-Kathrin Breyer	Social Medical Center – Otto Wagner Hospital, Vienna, Austria
	Georg Pall; Andreas Pircher	Universitätsklinik LKH Innsbruck, Innsbruck, Austria
	Horst Olschewski	Landeskrankenhaus – Universitätsklinik Graz, Graz, Austria
	Bernd Lamprecht	Keppler Universitätsklinikum, Linz, Austria
Brazil	Marcelo Malandrino de Albuquerque Felizola; Felipe Santa Rosa Roitberg	Instituto do Cancer do Estado de Sao Paulo, Sao Paulo, Brazil
	Maria Teresa Tsukazan; Carlos Henrique Escosteguy Barrios	Uniao Brasileira de Educacao e Assistencia Hospital Sao Lucas da PUCRS, Porto Alegre, Brazil
	Victor Marcondes Lopes dos Santos	Instituto Nacional Do Cancer Jose Alencar Gomes Da Silva, Rio de Janeiro, Brazil
	Thiago Bueno de Oliveira	A.C. Camargo Cancer Center, Sao Paulo, Brazil
	Mariana Ribeiro Monteiro	Hospital Paulistano - Amil Clinical Research, Sao Paulo, Brazil
	Katsuki Arima Tiscoski; Carlos Eugenio Escovar	Irmandade da Santa Casa de Misericordia de Porto Alegre, Porto Alegre, Brazil
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	Silvana Spadafora	Sault Area Hospital, Sault Ste Marie, ON
	Anthony Brade	Trillium Health Partners – Credit Valley Hospital, Mississauga, ON
	Paul Wheatley-Price	Ottawa Hospital Cancer Center, Ottawa, ON
	Mahmoud Abdelsalam	Moncton Hospital – Horizon Health Network, Moncton, NB
	Jean-Luc Dionne	CIUSSS de l'Est-de-l'Île-de-Montréal – Hôpital Maisonneuve-Rosemont, Montréal, QC
	Myriam Bouchard; Guy-Anne Turgeon	CHUS – Hôpital Fleurimont, Sherbrooke, QC
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	Helge Bischoff	Universitaetsklinikum Heidelberg, Heidelberg, Germany
	Thomas Wehler; Monika Serke; Alexander Baraniskin	Evangelisches Krankenhaus, Hamm, Germany
	Nikolaj Frost	Charite Universitaetsmedizin Berlin, Berlin, Germany
	Martin Faehling	Klinikum Esslingen-Klinik für Kardiologie und Pneumologie, Baden-Württemberg, Germany
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	Andrea Fulop	Orszagos Koranyi Pulmonologiai Intezet, Budapest, Hungary
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	Hiroshi Tanaka	Niigata Cancer Center Hospital, Niigata, Japan
	Takuya Shimizuguchi; Keiji Nihei	Tokyo Metropolitan Komagome Hospital, Tokyo, Japan
	Takeshi Kodaira	Aichi Cancer Center Hospital, Aichi, Japan
	Sojiro Kusumoto	Showa University Hospital, Tokyo, Japan
	Yuichi Takiguchi; Takuji Suzuki	Chiba University Hospital, Chiba, Japan
	Hiroshi Onishi	University of Yamanashi Hospital, Yamanashi, Japan
	Akito Hata	Kobe Minimally Invasive Cancer Center, Kobe, Japan
	Toru Kumagai; Motohiro Tamiya	Osaka International Cancer Institute, Osaka Japan
	Yasuhito Fujisaka	Osaka Medical and Pharmaceutical University Hospital, Osaka, Japan
	Shunichi Sugawara	Sendai Kousei Hospital, Sendai, Japan
	Koichi Azuma	Kurume University Hospital, Fukuoka, Japan
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Location	Current and Former Principal Investigator(s)	Site
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	Dmitry Bentsion	Sverdlovsk Regional Oncology Hospital, Ekaterinburg, Russia
	Konstantin Konstantinovich Laktionov	NN Blokhin Russian Cancer Research Center, Moscow, Russia
	Natalia Vladimirovna Fadeeva	Chelyabinsk Regional Clinical Oncology Dispensary, Chelyabinsk, Russia
South Korea	Myung-Ju Ahn	Samsung Medical Center, Seoul, South Korea
	Youngjoo Lee	National Cancer Center, Goyang-si, South Korea
	Ki Hyeong Lee	Chungbuk National University Hospital, Cheongju si, South Korea
	Byoung Yong Shim	The Catholic University of Korea, St Vincent s Hospital, Suwonsi, South Korea
Spain	Augusto Alejandro Valdivia Bustamante; Ana Callejo Pérez; Alex Martinez Marti	Hospital General Universitari Vall d'Hebron, Barcelona, Spain
	Ana Alvarez Gonzalez	Hospital General Universitario Gregorio Marañón, Madrid, Spain
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	Nicolas Mach; Franscesca Caparrotti; André Dante Durham	Hopitaux Universitaires de Genève HUG, Genève, Switzerland
Taiwan	Chen-Liang Tsai	Tri-Service General Hospital, Taipei, Taiwan
Turkey	Mustafa Erman	Hacettepe University School of Medicine, Oncology Hospital, Ankara, Turkey
	Umut Demirci; Cengiz Karacin; Berna Oksuzoglu	Oncology Training and Research Hospital, Ankara, Turkey
	Ozgur Ozyilkan	Başkent Adana Dr Turgut Noyan Uygulama ve Arastirma Merkezi, Adana, Turkey
	Mevlude Inanc	Erciyes University Medical Faculty, Kayseri, Turkey
	Ilhan Hacibekiroglu; Cemil Bilir	Sakarya Universitesi Tip Fakultesi Hastanesi, Korucuk, Turkey
	Cagatay Arslan	I.E.U. Medical Point Hastanesi, Izmir, Turkey
	Mahmut Gumus	Göztepe Prof. Dr. Süleyman Yalçın Şehir Hastanesi-oncology, Istanbul, Turkey
Ukraine	Viktoriya Leonova; Oleh Kobziev	Regional Center of Oncology-Thoracic Organs, Kharkiv, Ukraine
	Natalya Lisovska	Medical Center of Yuriy Spizhenko, Gliwice, Poland
	Stepan Krulko	Ukrainian Center of Tomotherapy, Kropyvnytskyi, Ukraine
The United Kingdom	Tony Pope; Anoop Haridass	Clatterbridge Cancer Center NHSFT, Bebington, UK
	Tathagata Das	Weston Park Hospital and Sheffield Teaching Hospitals NHSFT, Sheffield, UK
	Gareth Ayre	University Hospitals Bristol NHSFT, Bristol Haematology and Oncology Center, Bristol, UK
	Andreas Polychronis	Mount Vernon Hospital, Northwood, UK
	Eleni Karapanagiotou; Daniel Smith	Guy's and St Thomas' NHSFT, Guy's Cancer Center, London, UK
	Daniel Ajzensztejn	Oxford University Hospitals NHSFT, Oxford Cancer and Haematology Center, Oxford, UK
	Hemal Ariyaratne	Royal Free London NHSFT, London, UK

(continued on following page)

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TABLE A1. List of Current and Former Principal Study Investigators and Sites That Randomized One or More Participants (continued)

Location	Current and Former Principal Investigator(s)	Site
The United States	Laurie Carr	National Jewish Health, Denver, CO
	Roy H. Decker; Henry Park	Yale University, New Haven, CT
	Ebenezer Kio; James Wheeler	Goshen Center for Cancer Care, Goshen, IN
	Nadeem Ikhlaque; Nabin Khanal; Michael Eaton	Franciscan Health Indianapolis, Indianapolis, IN
	Susanne Markesbery Arnold	University of Kentucky School of Medicine & Hospitals, Lexington, KY
	Salma Jabbour	Rutgers Cancer Institute, New Brunswick, NJ
	Steven Powell; Ryan K Nowak	Sanford Cancer Center Oncology Clinic, Sioux Falls, SD
	Santosh Nair	Mid Florida Hematology and Oncology Center, Orange City, FL
	Leora Horn; Evan C. Osmundson	Vanderbilt University Medical Center, Nashville, TN
	Florence Keane	Massachusetts General Hospital, Boston, MA
	Wahid Hanna; Joseph Kelley	University of Tennessee Medical Center Knoxville, Knoxville, TN
	Patrick Cobb	Intermountain Health St Vincent Regional Hospital – Cancer Centers of Montana, Billings, MT
	Dan Costin	White Plains Hospital Center for Cancer Care, White Plains, NY
	Benny Weksler	Allegheny General Hospital, Pittsburgh, PA
	Millie Das	Veterans Affairs Palo Alto Health Care System, Palo Alto, CA
	Walter Sall	Mass General/North Shore Center for Outpatient Care, Danvers, MA
	Issam Makhoul	CARTI Cancer Center, Little Rock, AR
	Ajeet Gajra	Hematology Oncology Associates of Central New York, East Syracuse, NY

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TABLE A2. SBRT Intended Dose Regimen

Participants Treated With the Following SBRT Regimens	SBRT Plus Pembrolizumab (n = 226)[a]	SBRT Plus Placebo (n = 217)[a]
Intended dose regimen, No. (%)		
3 × 15 Gy	10 (4.4)	12 (5.5)
3 × 18 Gy	30 (13.3)	27 (12.4)
3 × 20 Gy	2 (0.9)	0 (0)
4 × 12 Gy	60 (26.5)	59 (27.2)
4 × 12.5 Gy	2 (0.9)	2 (0.9)
5 × 10 Gy	107 (47.3)	108 (49.8)
5 × 11 Gy	12 (5.3)	5 (2.3)
8 × 7.5 Gy	3 (1.3)	4 (1.8)

Abbreviation: SBRT, stereotactic body radiotherapy.
[a]Four participants were randomly assigned but not treated, and one participant was randomly assigned and treated with placebo but not SBRT because of lung collapse.

TABLE A3. Summary of AEs

Participants, No. (%)	SBRT Plus Pembrolizumab (n = 226)	SBRT Plus Placebo (n = 218)
With one or more AEs	216 (95.6)	200 (91.7)
Grade 3-5 AEs[a]	104 (46.0)	81 (37.2)
Serious AEs	88 (38.9)	71 (32.6)
Died	22 (9.7)[b]	15 (6.9)[c]
Discontinued treatment	60 (26.5)	25 (11.5)
Discontinued pembrolizumab or placebo	60 (26.5)	25 (11.5)
Discontinued radiation	1 (0.4)	0 (0.0)
Discontinued treatment because of a serious AE	37 (16.4)	19 (8.7)
Discontinued pembrolizumab or placebo	37 (16.4)	19 (8.7)
Discontinued radiation	0 (0.0)	0 (0.0)

Abbreviations: AE, adverse event; COPD, chronic obstructive pulmonary disease; NSCLC, non–small cell lung cancer; SBRT, stereotactic body radiotherapy.
[a]Grades were based on the National Cancer Institute Common Terminology Criteria for Adverse Events v4.03.
[b]Includes seven events of pneumonia, two events each of cardiac failure, COPD, death (not otherwise specified), and one event each of aortic aneurysm rupture, aspiration, bronchopneumopathy, COVID-19 pneumonia, gastric hemorrhage, Parkinson disease, peripheral arterial occlusive disease, pneumonitis, and sepsis.
[c]Includes two events each of COVID-19, cardiac arrest, and NSCLC recurrent, and one event each of COPD, COVID-19 pneumonia, cardiac failure congestive, cerebrovascular accident, colon cancer recurrent, drowning, renal failure, respiratory failure, and respiratory tract infection.

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TABLE A4. Causes of Death

Participants, No. (%)	SBRT Plus Pembrolizumab (n = 227)	SBRT Plus Placebo (n = 221)
Total dead	69 (30.4)	54 (24.4)
Malignant neoplasm progression	14 (6.2)	11 (5.0)
Pneumonia	11 (4.8)	2 (0.9)
Death	11 (4.8)	6 (2.7)
Cardiac arrest	4 (1.8)	3 (1.4)
Cardiac failure	3 (1.3)	2 (0.9)
Lung neoplasm malignant	3 (1.3)	2 (0.9)
Chronic obstructive pulmonary disease	3 (1.3)	1 (0.5)
COVID-19	2 (0.9)	2 (0.9)
Sepsis	2 (0.9)	1 (0.5)
NSCLC recurrent	1 (0.4)	5 (2.3)
Cerebrovascular accident	1 (0.4)	1 (0.5)
Multiple organ dysfunction syndrome	1 (0.4)	1 (0.5)
NSCLC	1 (0.4)	1 (0.5)
Aortic aneurysm rupture	1 (0.4)	0 (0)
Arteriosclerosis coronary artery	1 (0.4)	0 (0)
Bronchopneumopathy	1 (0.4)	0 (0)
Cardiovascular disorder	1 (0.4)	0 (0)
Cardiac failure chronic	1 (0.4)	0 (0)
Diverticular perforation	1 (0.4)	0 (0)
Gastric hemorrhage	1 (0.4)	0 (0)
Ischemia	1 (0.4)	0 (0)
Lung adenocarcinoma	1 (0.4)	0 (0)
Malnutrition	1 (0.4)	0 (0)
Parkinson disease	1 (0.4)	0 (0)
Pneumonitis	1 (0.4)	0 (0)
Respiratory failure	0 (0)	3 (1.4)
Assisted suicide	0 (0)	1 (0.5)
Cardiac failure congestive	0 (0)	1 (0.5)
Cardiopulmonary failure	0 (0)	1 (0.5)
Cardiovascular insufficiency	0 (0)	1 (0.5)
Colon cancer recurrent	0 (0)	1 (0.5)
Condition aggravated	0 (0)	1 (0.5)
COVID-19 pneumonia	0 (0)	1 (0.5)
Drowning	0 (0)	1 (0.5)
Dyspnea	0 (0)	1 (0.5)
Pneumonia aspiration	0 (0)	1 (0.5)
Renal failure	0 (0)	1 (0.5)
Respiratory disorder	0 (0)	1 (0.5)
Respiratory tract infection	0 (0)	1 (0.5)

Abbreviations: NSCLC, non–small cell lung cancer; SBRT, stereotactic body radiotherapy.

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TABLE A5. Summary of Pneumonitis and Radiation Pneumonitis AEs

Participants, No. (%)	SBRT Plus Pembrolizumab (n = 226)	SBRT Plus Placebo (n = 218)
With one or more AEs of pneumonitis or radiation pneumonitis	47 (20.8)	23 (10.6)
Grade 3-5 AEs	7 (3.1)	0 (0)

Participants, No. (%)	Grade 1 to 2	Grade 3 to 4	Grade 5	Grade 1 to 2	Grade 3 to 4	Grade 5
Injury, poisoning, and procedural complications	22 (9.7)	0 (0)	0 (0)	20 (9.2)	0 (0)	0 (0)
Radiation pneumonitis	22 (9.7)	0 (0)	0 (0)	20 (9.2)	0 (0)	0 (0)
Respiratory, thoracic, and mediastinal disorders	19 (8.4)	6 (2.7)	1 (0.4)	6 (2.8)	0 (0)	0 (0)
Immune-mediated lung disease	0 (0)	1 (0.4)	0 (0)	1 (0.5)	0 (0)	0 (0)
Interstitial lung disease	0 (0)	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)
Organizing pneumonia	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Pneumonitis	18 (8.0)	4 (1.8)	1 (0.4)	5 (2.3)	0 (0)	0 (0)

Abbreviations: AE, adverse event; SBRT, stereotactic body radiotherapy.

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TABLE A6. Immune-Mediated AEs and Infusion Reactions[a]

	SBRT Plus Pembrolizumab (n = 226)			SBRT Plus Placebo (n = 218)		
Type of AE	Grade 1 to 2, No. (%)	Grade 3 to 4, No. (%)	Grade 5, No. (%)	Grade 1 to 2, No. (%)	Grade 3 to 4, No. (%)	Grade 5, No. (%)
Hypothyroidism	36 (15.9)	0 (0)	0 (0)	8 (3.7)	0 (0)	0 (0)
Pneumonitis	19 (8.4)	6 (2.7)	1 (0.4)	6 (2.8)	0 (0)	0 (0)
Hyperthyroidism	16 (7.1)	0 (0)	0 (0)	5 (2.3)	0 (0)	0 (0)
Colitis	2 (0.9)	3 (1.3)	0 (0)	0 (0)	1 (0.5)	0 (0)
Severe skin reactions	0 (0)	4 (1.8)	0 (0)	0 (0)	0 (0)	0 (0)
Hepatitis	0 (0)	3 (1.3)	0 (0)	0 (0)	0 (0)	0 (0)
Pancreatitis	0 (0)	2 (0.9)	0 (0)	0 (0)	0 (0)	0 (0)
Adrenal insufficiency	0 (0)	1 (0.4)	0 (0)	4 (1.8)	0 (0)	0 (0)
Arthritis	1 (0.4)	0 (0)	0 (0)	1 (0.5)	0 (0)	0 (0)
Encephalitis	0 (0)	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)
Exocrine pancreatic insufficiency	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Gastritis	1 (0.4)	0 (0)	0 (0)	1 (0.5)	0 (0)	0 (0)
Infusion reactions	0 (0)	1 (0.4)	0 (0)	1 (0.5)	1 (0.5)	0 (0)
Myocarditis	0 (0)	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)
Myositis	1 (0.4)	0 (0)	0 (0)	1 (0.5)	0 (0)	0 (0)
Nephritis	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Sarcoidosis	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Thyroiditis	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Uveitis	1 (0.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)

Abbreviations: AE, adverse event; SBRT, stereotactic body radiotherapy.
[a]Immune-mediated AEs and infusion reactions were based on a list of preferred terms intended to capture known risks of pembrolizumab and were considered regardless of attribution to study treatment by the investigator.

TABLE A7. Analyses of Change From Baseline to Week 24 in Participant-Reported Outcome Measures

PRO Measure	N	Baseline Mean (SD)	N	Week 24 Mean (SD)	N	Least-squares mean change (95% CI)[b] from baseline to week 24
EORTC QLQ-C30[a] Global Health Status						
SBRT plus pembrolizumab	220	64.66 (21.10)	172	63.23 (20.57)	225	−2.17 (−5.00 to 0.66)
SBRT plus placebo	204	65.28 (19.22)	177	65.21 (18.50)	214	−0.96 (−3.78 to 1.86)
Difference[c] in least-squares means[b] (95% CI); *P* value[b]						−1.21 (−4.90 to 2.48); *P* = .52
EORTC QLQ-C30[a] Physical Functioning						
SBRT plus pembrolizumab	220	72.70 (21.90)	172	73.22 (21.41)	225	−0.52 (−3.08 to 2.03)
SBRT plus placebo	204	72.65 (20.12)	177	73.79 (20.97)	214	−0.31 (−2.86 to 2.24)
Difference[c] in least-squares means[b] (95% CI); *P* value[b]						−0.21 (−3.70 to 3.28); *P* = .91
EORTC QLQ-C30[a] Role Functioning						
SBRT plus pembrolizumab	220	75.83 (28.77)	172	72.29 (29.74)	225	−4.15 (−7.92 to −0.37)
SBRT plus placebo	204	75.08 (26.85)	177	77.12 (28.13)	214	−0.04 (−3.80 to 3.72)
Difference[c] in least-squares means[b] (95% CI); *P* value[b]						−4.10 (−9.21 to 1.00); *P* = .11
EORTC QLQ-LC13[d] Chest Pain						
SBRT plus pembrolizumab	219	8.37 (17.37)	170	10.39 (21.83)	225	3.00 (−0.09 to 6.10)
SBRT plus placebo	201	6.80 (16.44)	175	8.00 (17.12)	214	1.76 (−1.32 to 4.84)
Difference[c] in least-squares means[b] (95% CI); *P* value[b]						1.24 (−2.83 to 5.31); *P* = .55
EORTC QLQ-LC13[d] Coughing						
SBRT plus pembrolizumab	219	27.25 (24.18)	170	26.08 (26.27)	225	−1.44 (−5.23 to 2.36)
SBRT plus placebo	201	30.35 (27.32)	175	31.05 (26.64)	214	2.91 (−0.87 to 6.68)
Difference[c] in least-squares means[b] (95% CI); *P* value[b]						−4.34 (−9.38 to 0.69); *P* = .09

Abbreviations: ECOG PS, Eastern Cooperative Oncology Group performance status; EORTC QLQ-C30 or -LC13, European Organization for the Research and Treatment of Cancer Quality of Life Questionnaire-Core 30 or -Lung Cancer 13; PRO, participant-reported outcomes; SBRT, stereotactic body radiotherapy; SD, standard deviation.

[a]A higher score for EORTC QLQ-C30 Global Health Status, Physical Functioning, and Role Functioning indicates relatively better global health status, physical functioning, and role functioning than a lower score.

[b]Based on a constrained longitudinal data analysis model with PRO scores as the response variable with covariates for treatment by study visit interaction, and stratification factors by disease stage (I *v* II), ECOG performance status (0 or 1 *v* 2), geographic region (east Asia *v* non-East Asia), and reason for not receiving surgery (medically inoperable *v* refused surgery), with small strata collapsed. *P* value is two-sided.

[c]Difference between the SBRT plus pembrolizumab and SBRT plus placebo arms.

[d]A higher score for EORTC QLQ-LC13 single items indicates relatively more severe symptoms than a lower score.

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